ODE Definitions and Terminology

- A **first-order linear IVP** is $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{f}(t)$ with $\boldsymbol{y}(0) = \boldsymbol{y}_0$.
  - The **general solution (GS)** is $\boldsymbol{y}(t) = \boldsymbol{y}_c(t) + \boldsymbol{y}_p(t)$.
  - The **complementary solution** $\boldsymbol{y}_c$ (containing constants $\boldsymbol{c}$) satisfies $\boldsymbol{y}_c' = \boldsymbol{A}(t)\boldsymbol{y}_c + \boldsymbol{0}$.
  - A **particular solution** $\boldsymbol{y}_p$ (no undetermined constants) satisfies $\boldsymbol{y}_p' = \boldsymbol{A}(t)\boldsymbol{y}_p + \boldsymbol{f}(t)$.
  - The constants $\boldsymbol{c}$ in the GS are determined by solving the LS $\boldsymbol{y}_c(0) + \boldsymbol{y}_p(0) = \boldsymbol{y}_0$.
- **Constant Coefficient** means $\boldsymbol{A}$ is independent of t.
  - Eigenpairs satisfying $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ give solutions $e^{\lambda_i t}\boldsymbol{v}_i$ of $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$
  - With a complete LI set of eigenvectors $\boldsymbol{y}_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t}\boldsymbol{v}_i$.
  - **Undetermined Coefficients** guesses a $\boldsymbol{y}_p$ form by looking at $\boldsymbol{f}(t)$. The coefficients are determined by substituting into the ODE $\boldsymbol{y}_p' = \boldsymbol{A}(t)\boldsymbol{y}_p + \boldsymbol{f}(t)$.

Exponential Forcing

$$\boldsymbol{y}' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{\boldsymbol{A}} \boldsymbol{y} + \overbrace{\begin{pmatrix} 7 \\ 2 \end{pmatrix} e^{2t}}^{\boldsymbol{f}(t)}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a}e^{2t}$ which gives $\boldsymbol{y}_p' = 2\boldsymbol{a}e^{2t}$ in the ODE

$$2\boldsymbol{a}e^{2t} = \boldsymbol{A}\boldsymbol{a}e^{2t} + \begin{pmatrix} 7 \\ 2 \end{pmatrix} e^{2t} \quad \Longrightarrow \quad (2\boldsymbol{I} - \boldsymbol{A})\boldsymbol{a} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -1 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix} \quad \Longrightarrow \quad \left[\begin{array}{cc|c} 4 & -1 & 7 \\ -1 & 4 & 2 \end{array}\right] \sim \left[\begin{array}{cc|c} 4 & -1 & 7 \\ 0 & 15 & 15 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

- Thus $\boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\boldsymbol{y}_p(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t}$.
- General solution

$$\boldsymbol{y}(t) = \boldsymbol{y}_p(t) + \boldsymbol{y}_c(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + c_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- As before, initial conditions $\boldsymbol{y}(0) = \boldsymbol{y}_0$ determine the constants.

Exponential Forcing

$$y' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{\boldsymbol{A}} \boldsymbol{y} + \overbrace{\begin{pmatrix} 7 \\ 2 \end{pmatrix} e^{-4t}}^{\boldsymbol{f}(t)}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a}e^{-4t}$ which gives $\boldsymbol{y}_p' = -4\boldsymbol{a}e^{-4t}$ in the ODE

$$-4\boldsymbol{a}e^{-4t} = \boldsymbol{A}\boldsymbol{a}e^{-4t} + \begin{pmatrix} 7 \\ 2 \end{pmatrix} e^{-4t} \quad \Longrightarrow \quad (-4\boldsymbol{I} - \boldsymbol{A})\boldsymbol{a} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix} \quad \Longrightarrow \quad \left[\begin{array}{cc|c} -2 & -1 & 7 \\ -1 & -2 & 2 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 3 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & -4 \\ 0 & 1 & 1 \end{array}\right]$$

- Thus $\boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ and $\boldsymbol{y}_p(t) = \begin{pmatrix} -4 \\ 1 \end{pmatrix} e^{-4t}$.
- General solution

$$\boldsymbol{y}(t) = \boldsymbol{y}_p(t) + \boldsymbol{y}_c(t) = \begin{pmatrix} -4 \\ 1 \end{pmatrix} e^{-4t} + c_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- As always, initial conditions $\boldsymbol{y}(0) = \boldsymbol{y}_0$ determine the constants.

Sinusoidal Forcing

$$\mathbf{y}' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{\mathbf{A}} \mathbf{y} + \overbrace{\begin{pmatrix} 8 \\ -2 \end{pmatrix} \sin(t)}^{\mathbf{f}(t)}$$

- Try $\mathbf{y}_p(t) = \mathbf{a}\sin(t) + \mathbf{b}\cos(t)$ which gives $\mathbf{y}_p'(t) = \mathbf{a}\cos(t) - \mathbf{b}\sin(t)$ in the ODE

$$\mathbf{a}\cos(t) - \mathbf{b}\sin(t) = \mathbf{A}\mathbf{a}\sin(t) + \mathbf{A}\mathbf{b}\cos(t) + \begin{pmatrix} 8 \\ -2 \end{pmatrix}\sin(t)$$

Match sine and cosine terms

$$\mathbf{A}\mathbf{a} + \mathbf{b} = -\begin{pmatrix} 8 \\ -2 \end{pmatrix} \qquad \mathbf{a} - \mathbf{A}\mathbf{b} = \mathbf{0} \quad \Longrightarrow \quad \begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & -\mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} = \begin{pmatrix} -\begin{pmatrix} 8 \\ -2 \end{pmatrix} \\ \mathbf{0} \end{pmatrix}$$

$$\left[\begin{array}{cc|cc|c} -2 & 1 & 1 & 0 & -8 \\ 1 & -2 & 0 & 1 & 2 \\ \hline 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -1 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right] \quad \Longrightarrow \quad \mathbf{a} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \qquad \mathbf{b} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

- General solution $\mathbf{y}(t) = \begin{pmatrix} 3 \\ 0 \end{pmatrix}\sin(t) + \begin{pmatrix} -2 \\ -1 \end{pmatrix}\cos(t) + c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Sinusoidal Forcing

$$\boldsymbol{y}' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{\boldsymbol{A}} \boldsymbol{y} + \overbrace{\begin{pmatrix} 5\sin(2t) \\ -4\sin(2t) + 7\cos(2t) \end{pmatrix}}^{\boldsymbol{f}(t)}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a}\sin(2t) + \boldsymbol{b}\cos(2t)$ which gives $\boldsymbol{y}_p'(t) = 2\boldsymbol{a}\cos(2t) - 2\boldsymbol{b}\sin(2t)$ in the ODE

$$2\boldsymbol{a}\cos(2t) - 2\boldsymbol{b}\sin(2t) = \boldsymbol{A}\boldsymbol{a}\sin(2t) + \boldsymbol{A}\boldsymbol{b}\cos(2t) + \begin{pmatrix} 5 \\ -4 \end{pmatrix}\sin(2t) + \begin{pmatrix} 0 \\ 7 \end{pmatrix}\cos(2t)$$

Match sine and cosine terms

$$\boldsymbol{A}\boldsymbol{a} + 2\boldsymbol{b} = -\begin{pmatrix} 5 \\ -4 \end{pmatrix} \qquad 2\boldsymbol{a} - \boldsymbol{A}\boldsymbol{b} = \begin{pmatrix} 0 \\ 7 \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} \boldsymbol{A} & 2\boldsymbol{I} \\ 2\boldsymbol{I} & -\boldsymbol{A} \end{pmatrix}\begin{pmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{pmatrix} = \begin{pmatrix} -5 \\ 4 \\ 0 \\ 7 \end{pmatrix}$$

$$\left[\begin{array}{cc|cc|c} -2 & 1 & 2 & 0 & -5 \\ 1 & -2 & 0 & 2 & 4 \\ \hline 2 & 0 & 2 & -1 & 0 \\ 0 & 2 & -1 & 2 & 7 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] \quad \Longrightarrow \quad \boldsymbol{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

- General solution $\boldsymbol{y}(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}\sin(2t) + \begin{pmatrix} -1 \\ 2 \end{pmatrix}\cos(2t) + c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Superposition

$$y' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{A} y + \overbrace{\begin{pmatrix} 7 \\ 2 \end{pmatrix} e^{-4t}}^{f_1(t)} + \overbrace{\begin{pmatrix} 5\sin(2t) \\ -4\sin(2t) + 7\cos(2t) \end{pmatrix}}^{f_2(t)}$$

- We already found particular solutions for each forcing function

$$y_{p,1}(t) = \begin{pmatrix} -4 \\ 1 \end{pmatrix} e^{-4t} \qquad y_{p,2}(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \sin(2t) + \begin{pmatrix} -1 \\ 2 \end{pmatrix} \cos(2t)$$

- Their sum $y_p = y_{p,1} + y_{p,2}$ is a particular solution because

$$y_p' = y_{p,1}' + y_{p,2}' = Ay_{p,1} + f_1 + Ay_{p,2} + f_2 = A\underbrace{(y_{p,1} + y_{p,2})}_{y_p} + f_1 + f_2$$

- General solution

$$y(t) = \begin{pmatrix} -4 \\ 1 \end{pmatrix} e^{-4t} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} \sin(2t) + \begin{pmatrix} -1 \\ 2 \end{pmatrix} \cos(2t) + c_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Undetermined Coefficients $y' = Ay + f(t)$ constant $A \in \mathbb{R}^{n\times n}$

Compute $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ for $i = 1, \dots n$ and write down

$$\boldsymbol{y}_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} \boldsymbol{v}_i$$

Choose $\boldsymbol{y}_p$ form and determine coefficients vectors by substituting into the ODE.

- **Exponentials** $\boldsymbol{f}(t) = \boldsymbol{u}e^{\lambda t}$. Try $\boldsymbol{y}_p(t) = \boldsymbol{a}e^{\lambda t}$.
- **Sines and cosines** $\boldsymbol{f}(t) = \boldsymbol{u}\sin(\omega t) + \boldsymbol{v}\cos(\omega t)$. Try $\boldsymbol{y}_p(t) = \boldsymbol{a}\sin(\omega t) + \boldsymbol{b}\cos(\omega t)$.
  - Include both sine and cosine even when only one appears in the forcing.
- **Polynomials** $\boldsymbol{f}(t) = \sum_{k=0}^{m} \boldsymbol{u}_k t^k$. Try $\boldsymbol{y}_p(t) = \sum_{k=0}^{m} \boldsymbol{a}_k t^k$.
  - Include every power from t^m down to the constant term.
- **Superposition** If $\boldsymbol{f} = \boldsymbol{f}_1 + \boldsymbol{f}_2$ add particular solutions $\boldsymbol{y}_p = \boldsymbol{y}_{p,1} + \boldsymbol{y}_{p,2}$.

General solution $\boldsymbol{y} = \boldsymbol{y}_c + \boldsymbol{y}_p$. What can go wrong?